

SYDNEY TECHNICAL HIGH SCHOOL



PRELIMINARY HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 1

MAY 2014

Mathematics Extension 1

General Instructions

- Working time - 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- In questions 6 to 11, show relevant mathematical reasoning and/or calculations
- Start each question in section 2 on a new page

Total marks - 53

Section 1 - 5 marks

Attempt Questions 1 – 5.
Allow about 7 minutes for this section.

Section 2 - 48 marks

Attempt Questions 6 – 11.
Allow about 63 minutes for this section.

Name : _____

Teacher : _____

Section 1

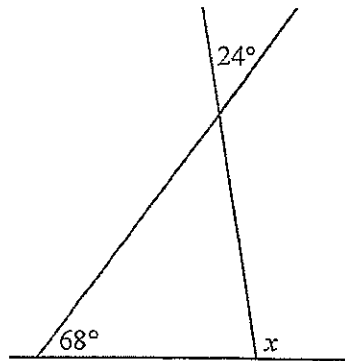
5 marks

Attempt Questions 1 – 5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 5.
Do not remove the multiple-choice answer sheet from your answer booklet.

1



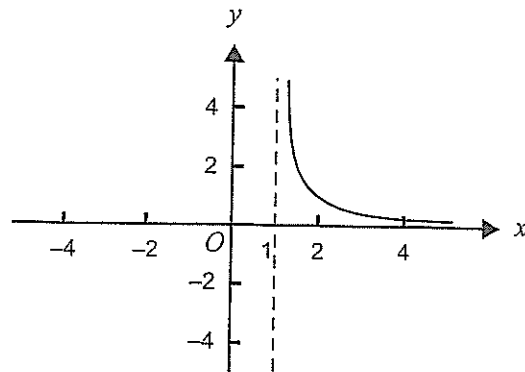
The size of angle x is

- A. 88°
- B. 92°
- C. 112°
- D. 116°

2. How many asymptotes does the graph of the function $y = \frac{3x^2}{x(2-x)}$ have?

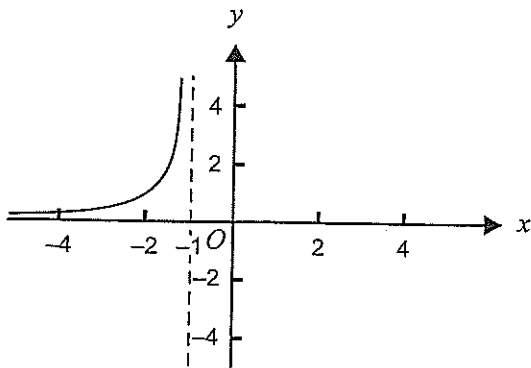
- A. 0
- B. 1
- C. 2
- D. 3

3. Part of the graph of the function with rule $y = f(x)$ is shown below

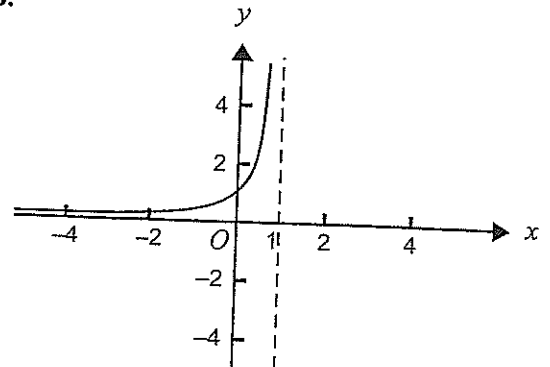


Which one of the following is most likely to be the corresponding part of the function with rule $y = f(-x)$?

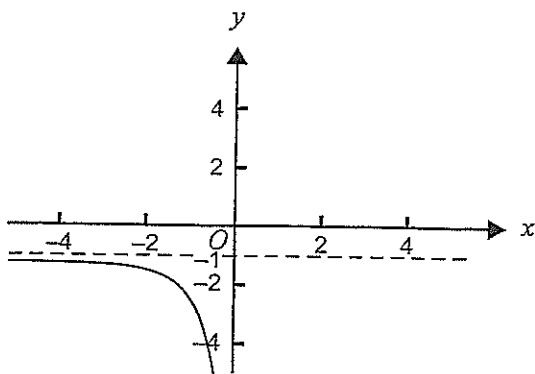
A.



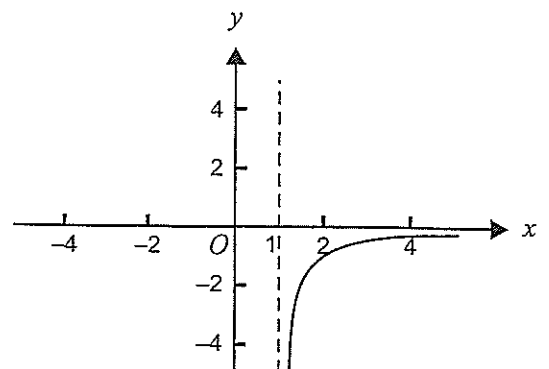
B.



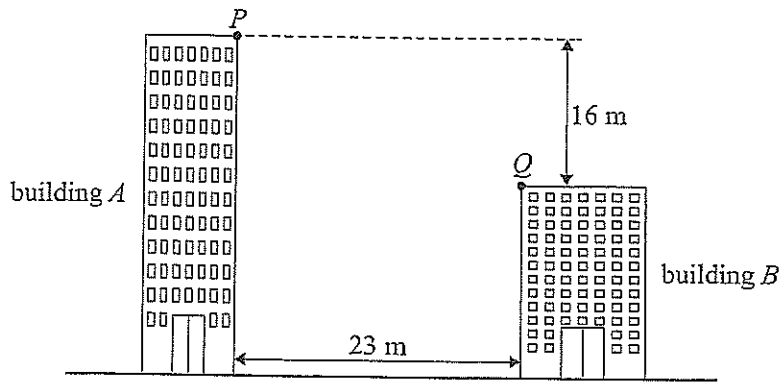
C.



D.



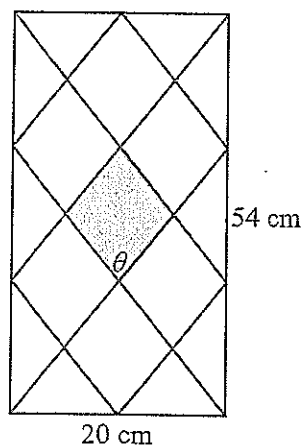
4.



In the diagram above, the angle of depression of point Q from point P is closest to

- A. 35°
- B. 41°
- C. 46°
- D. 55°

5. The rectangle shown below is 54 cm high and 20 cm wide.
The rhombuses drawn inside the rectangle are all the same size and shape.



The size of the angle θ , in the shaded rhombus, is closest to

- A. 34°
- B. 56°
- C. 58°
- D. 67°

Section 2

48 marks

Attempt Questions 6 – 11

Allow about 63 minutes for this section

Answer each question in your answer booklet. Start each question on a new page.

In Questions 6 – 11, your response should include relevant mathematical reasoning and/or calculations.

Question 6 (8 marks)

- a) Fully factorise $2x^4 + 16x$ 1
- b) If $\sec \theta = 3$ and $\tan \theta < 0$, find the exact value of $\sin \theta$. 2
- c) Simplify $\left(\frac{x^{p+q}}{x^q}\right)^p \div \left(\frac{x^q}{x^{q-p}}\right)^{p-q}$ 2
- d) If $g(5x) = 50x^2 + 10x + 1$, find an expression for $g(x)$. 1
- e) Draw a neat sketch of $y = \frac{x-2}{x+2}$ 2

Question 7 (8 marks) Start a new page

- a) Solve $\sin(\theta - 75^\circ) = \frac{-\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$ 2
- b) If A is an acute angle, simplify $\frac{\tan A}{\sqrt{1+\tan^2 A}}$ 2
- c) In pentagon $ABCDE$, angle $A = 120^\circ$, angle $E = 140^\circ$,
 AB is parallel to DC , and BC is parallel to AE .
- i) Draw a neat sketch clearly showing this information. 1
- ii) Find the size of angle B , giving reasons. 1
- iii) Find the size of angle D , giving reasons. 2

Question 8 (8 marks) Start a new page

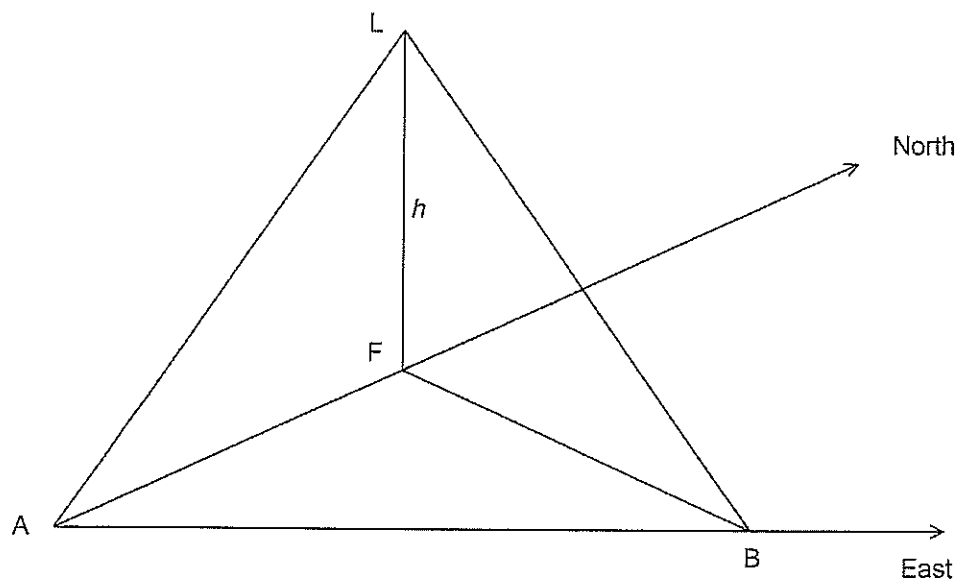
- a) Solve $2\cos^2 x = \sin x + 1$, for $0^\circ \leq x \leq 360^\circ$. 3
- b) Simplify $\frac{5^{-n} \times 25^{2n-2}}{5^{3n-2} \times 10^{-1}}$ 2
- c) Solve $\frac{5}{4-x} \geq 1$ 3

Question 9 (8 marks) Start a new page

a) Solve $|2x - 1| = 3x + 6$ 2

b) Show that $\sec \alpha - \cos \alpha = \sin \alpha \tan \alpha$ 2

c)



A vertical flagpole, FL , of height h metres stands in the middle of a park. From point A , due South of the flagpole, the angle of elevation to the top of the flagpole is 35° . From point B , which is 45 metres due East of point A , the angle of elevation to the top of the flagpole is 28° .

- i) Find an expression for the length of AF in terms of h . 1
- ii) Find the height of the flagpole, in metres correct to 1 decimal place. 3

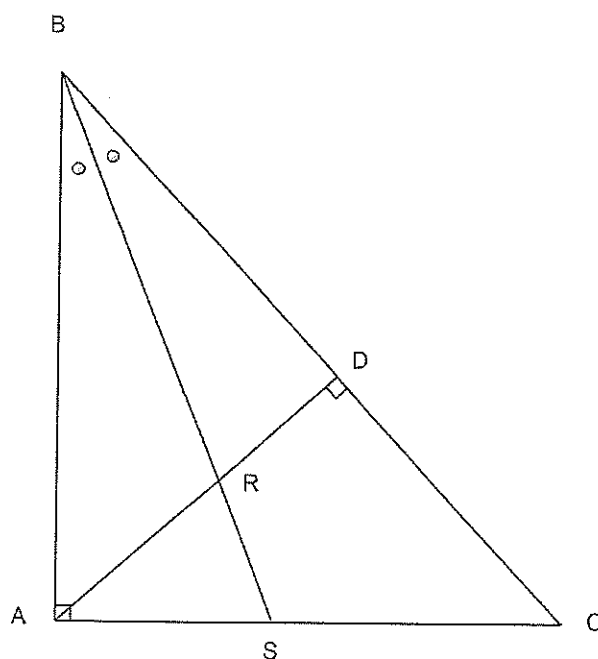
Question 10 (8 marks) Start a new page

a) Solve $(2x - 1)^2 = 5$ 1

b) i) Draw a neat sketch of $y = x^2 - 6x + 8$, 2
clearly showing all intercepts and the vertex.

ii) On a separate diagram draw a neat sketch of $y = \frac{1}{x^2 - 6x + 8}$ 2
clearly showing all important features.

c)



In triangle ABC, angle $A = 90^\circ$, SB bisects angle B and AD is perpendicular 3
to BC and meets SB at R .

By letting angle $SBC = x$, or otherwise, prove that triangle ASR is isosceles.

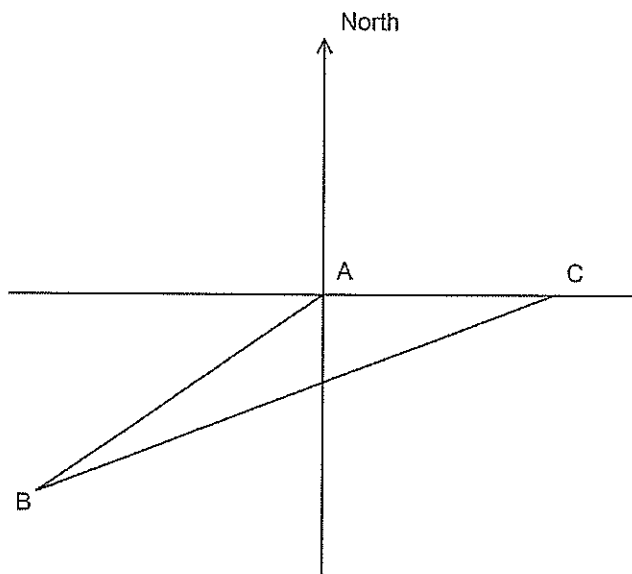
Question 11 (8 marks) Start a new page

- a) Solve simultaneously for x and y ,

2

$$y = x^2 - 2x - 1 \quad \text{and} \quad 2x - y - 1 = 0$$

- b)



A surveyor standing at point A notes that, point B is on a bearing of $228^\circ T$ and point C is due East of point A . The surveyor then walks 85 metres to point B where he notes that the bearing of point C from point B is $070^\circ T$.

3

Find the distance from point B to point C .
(Give answer in metres correct to 1 decimal place)

- c) Solve $|x + 1| > \sqrt{25 - x^2}$

3

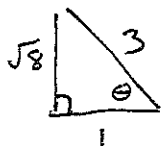
End of paper

Ext 1 SOLUTIONS MAY 2014

1. B
2. D
3. A
4. A
5. C

6. a) $2x(x+2)(x^2-2x+4)$

b) 4th quad



$$\sin \theta = -\frac{\sqrt{8}}{3}$$

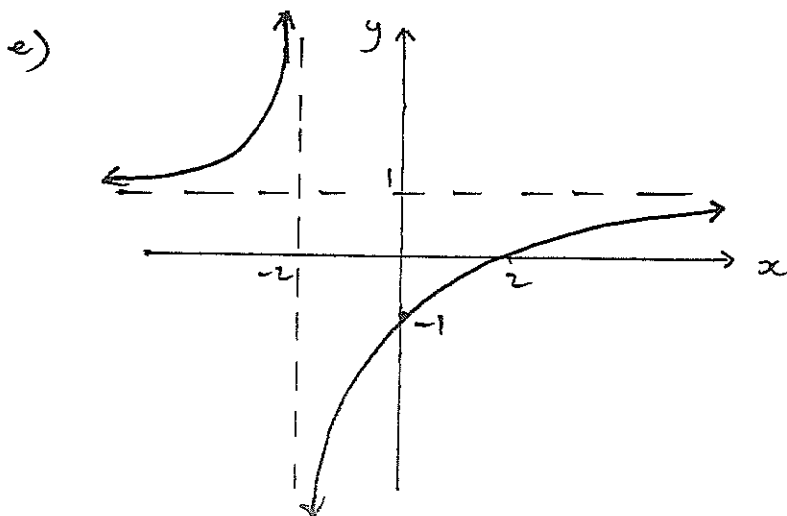
c) $\left(\frac{x^{p+q}}{x^q}\right)^p \div \left(\frac{x^q}{x^{q-p}}\right)^{p-q}$

$$= (x^p)^p \div (x^p)^{p-q}$$

$$= x^{pq}$$

d) $g(5x) = 2(5x)^2 + 2(5x) + 1$

$$\therefore g(x) = 2x^2 + 2x + 1$$



7.

$$a. \quad \theta - 75^\circ = 240^\circ, 300^\circ$$

$$\theta = 315^\circ, 375^\circ$$

$$\therefore \theta = 315^\circ, 15^\circ$$

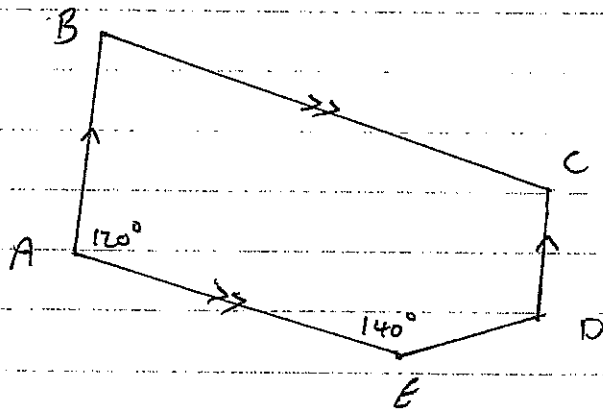
$$b. \quad \frac{\tan A}{\sqrt{1+\tan^2 A}}$$

$$= \frac{\tan A}{\sqrt{\sec^2 A}}$$

$$= \tan A \cos A$$

$$= \sin A$$

c. i)



$$ii) \quad \angle B = 60^\circ \text{ (co-interior angles, } BC \parallel AE)$$

$$iii) \quad \angle C = 120^\circ \text{ (co-interior angles, } AB \parallel EC)$$

$$\text{angle sum of pentagon} = 540^\circ$$

$$\therefore \angle D = 540^\circ - 120^\circ - 140^\circ - 60^\circ - 120^\circ$$

$$= 100^\circ$$

8.

$$\begin{aligned}
 \text{a.} \quad & 2 \cos^2 x = \sin x + 1 \\
 & 2(1 - \sin^2 x) = \sin x + 1 \\
 & 2 \sin^2 x + \sin x - 1 = 0 \\
 & (2 \sin x - 1)(\sin x + 1) = 0 \\
 & \sin x = \frac{1}{2}, -1
 \end{aligned}$$

$$x = 30^\circ, 150^\circ, 270^\circ$$

$$\begin{aligned}
 \text{b.} \quad & \frac{5^{-n} \times 25^{2n-2}}{5^{3n-2} \times 10^{-1}} \\
 & = \frac{5^{-n} \times (5^2)^{2n-2}}{5^{3n-2} \times 5^{-1} \times 2^{-1}} \\
 & = \frac{5^{-n} \times 5^{4n-4}}{5^{3n-2} \times 5^{-1} \times 2^{-1}} \\
 & = \frac{5^{3n-4}}{5^{3n-3} \times 2^{-1}} \\
 & = 5^{-1} \times 2 \\
 & = \frac{2}{5}
 \end{aligned}$$

$$\text{c.} \quad \frac{5}{4-x} \geq 1$$

$$\frac{5(4-x)^2}{4-x} \geq (4-x)^2$$

$$5(4-x) \geq (4-x)^2$$

$$(4-x)^2 - 5(4-x) \leq 0$$

$$(4-x)(-x-1) \leq 0$$

$$-1 \leq x < 4$$

9.

a. $|2x-1| = 3x+6$

$$2x-1 = 3x+6$$

$$x = -7$$

test $x = -7$ ✗

$$2x-1 = -3x-6$$

$$5x = -5$$

$$x = -1$$

$x = -1$ ✓

$\therefore x = -1$

b. LHS = $\sec \alpha - \cos \alpha$

$$= \frac{1}{\cos \alpha} - \cos \alpha$$

$$= \frac{1 - \cos^2 \alpha}{\cos \alpha}$$

$$= \frac{\sin^2 \alpha}{\cos \alpha}$$

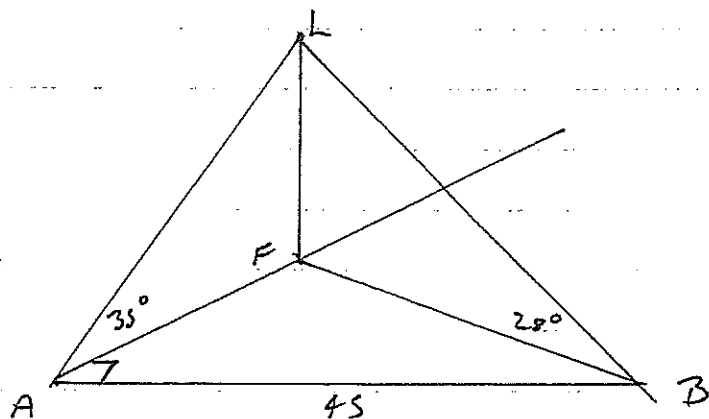
$$= \sin \alpha \tan \alpha$$

$$= \text{RHS}$$

c. i. $\tan 35^\circ = \frac{h}{AF}$

$$AF = \frac{h}{\tan 35^\circ}$$

$$= h \tan 55^\circ$$



ii. $BF = h \tan 62^\circ$

$$AB^2 + AF^2 = BF^2$$

$$45^2 = h^2 \tan^2 62^\circ - h^2 \tan^2 55^\circ$$

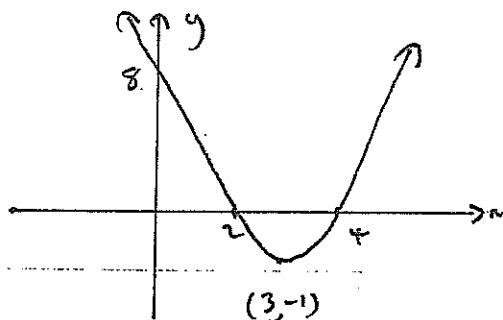
$$h = \frac{45}{\sqrt{\tan^2 62^\circ - \tan^2 55^\circ}}$$

$$= 36.8 \text{ m.}$$

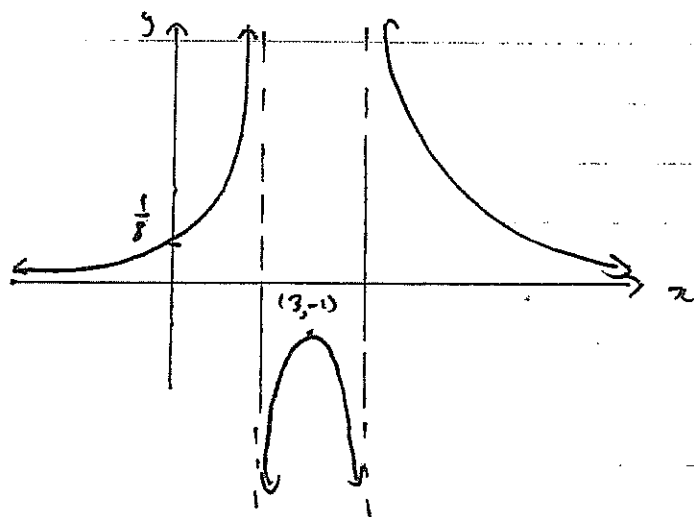
10.

a. $(2x-1)^2 = 5$
 $2x-1 = \pm\sqrt{5}$
 $x = \frac{1 \pm \sqrt{5}}{2}$

b. i. $y = (x-4)(x-2)$



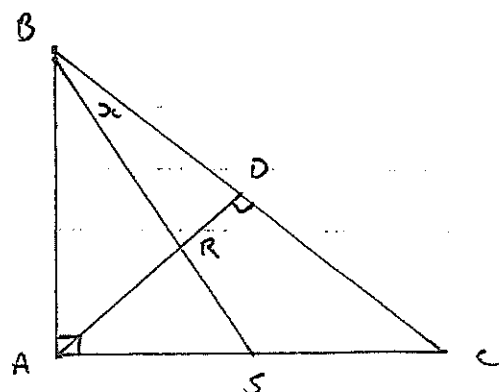
ii.



c. Let $\angle SBC = x$
 $\angle BRD = 90 - x$ (angle sum of $\triangle BDR$)
 $\angle ARS = 90 - x$ (vertically opposite)
 $\angle ABS = x$ (equal to $\angle SBC$)
 $\therefore \angle ASB = 90 - x$ (angle sum of $\triangle ABS$)

$\therefore \angle ARS = \angle ASB$

$\therefore ASR$ is isosceles



11.

$$a. \quad y = x^2 - 2x - 1$$

$$2x - y - 1 = 0 \quad \xrightarrow{\text{sub}} \quad y = 2x - 1$$

$$2x - 1 = x^2 - 2x - 1$$

$$x^2 - 4x = 0$$

$$x(x - 4) = 0$$

$$x = 0, 4$$

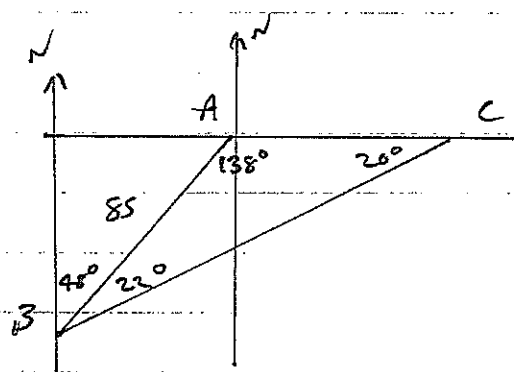
$$\therefore y = -1, 7$$

$\therefore$ solutions $x = 0, y = -1$ and $x = 4, y = 7$

$$b. \quad \frac{BC}{\sin 138^\circ} = \frac{85}{\sin 20^\circ}$$

$$BC = \frac{85 \times \sin 138^\circ}{\sin 20^\circ}$$

$$= 166.3 \text{ m.}$$



$$c. \quad y = |x+1|$$

$$y = \sqrt{25-x^2}$$

$$x+1 = \sqrt{25-x^2}$$

$$(x+1)^2 = 25-x^2$$

$$2x^2 + 2x - 24 = 0$$

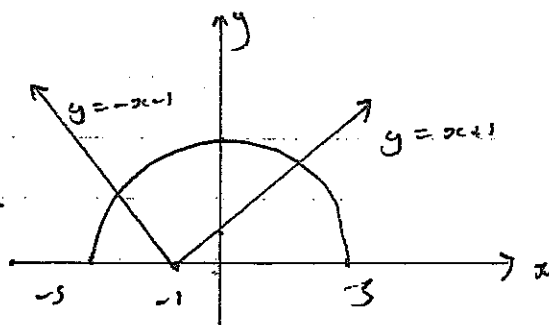
$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

$$x = -4, 3$$

$$-x+1 = \sqrt{25-x^2}$$

same



$$\therefore -5 \leq x < -4, \quad 3 < x \leq 5$$